LT 2
Conditional

Conditional

- An `if-elif-else` structure works like a **sequence of gates**.
- Conditions are checked **in order, from top to bottom**.
- The **first condition that is true** allows execution to pass through its gate.
  - Once a gate is taken, **all remaining conditions are skipped**.
- If **no conditions are true**, the else block (if present) handles the leftover case.
- The else block is **optional** — it may do something or nothing.

Key idea: At most **one block** of code runs in an `if-elif-else` structure.

Conditional

Form I

```
if expr:
        statement(s)
```

indentation

e.g.

```
if a > 0:
    print('a is positive')

a = 3

>>> 'a is positive'
```

This is the output.

Indentation Matters !

This is a Python thingy !!!

Conditional

Form 2

```
if expr:
    statement(s)
else:
    statement(s)
```

indentation

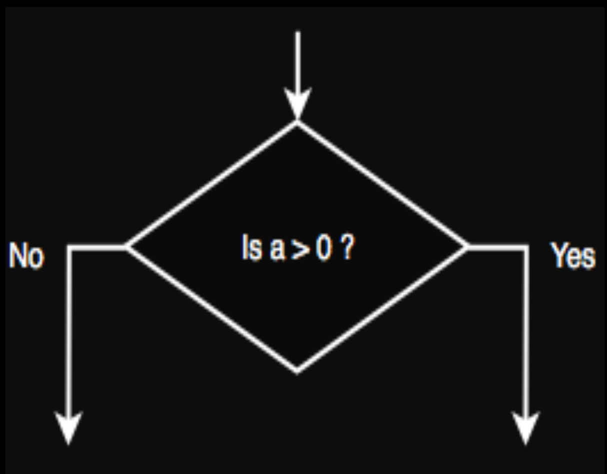
e.g.

```
if a > 0:
    print('a is positive.')
else:
    print('a is not positive.')

a = -3

>>> 'a is not positive.'
```

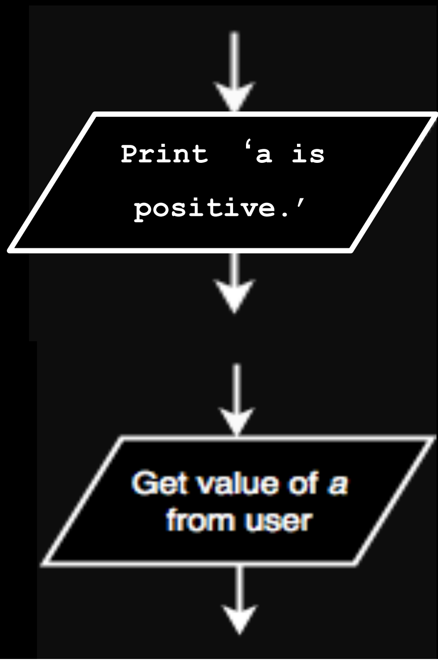
Flowcharts

(not in syllabus)

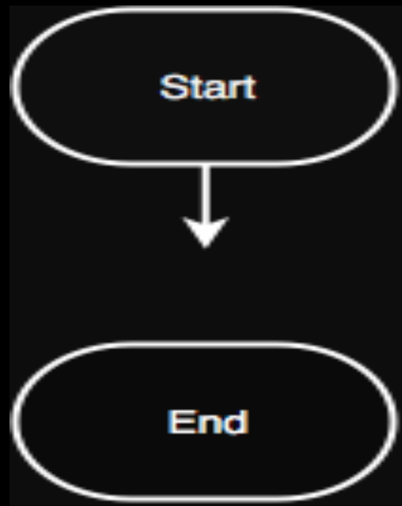
Decision Nodes



Input / Output Nodes

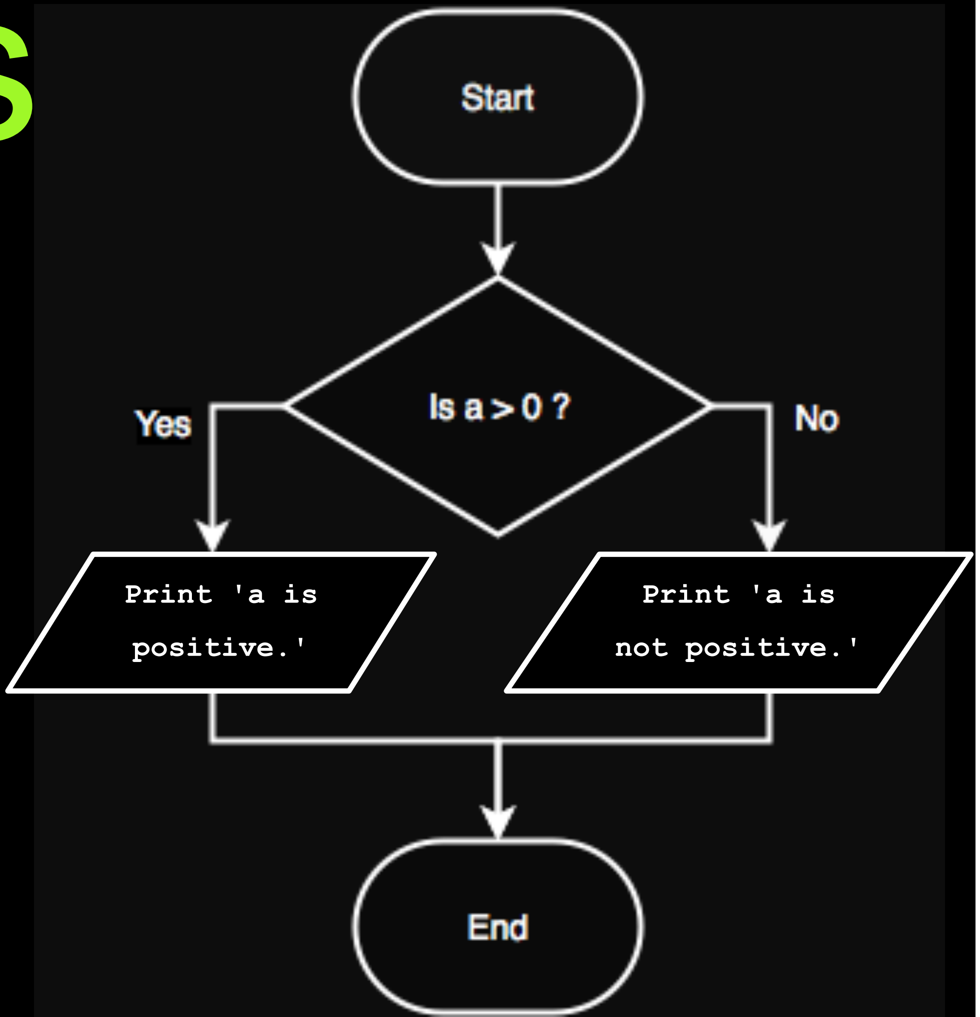


Terminal Nodes



Flowcharts

```
if a > 0:
     print('a is positive.')
else:
     print('a is not positive.')
```



Conditional

Form 3

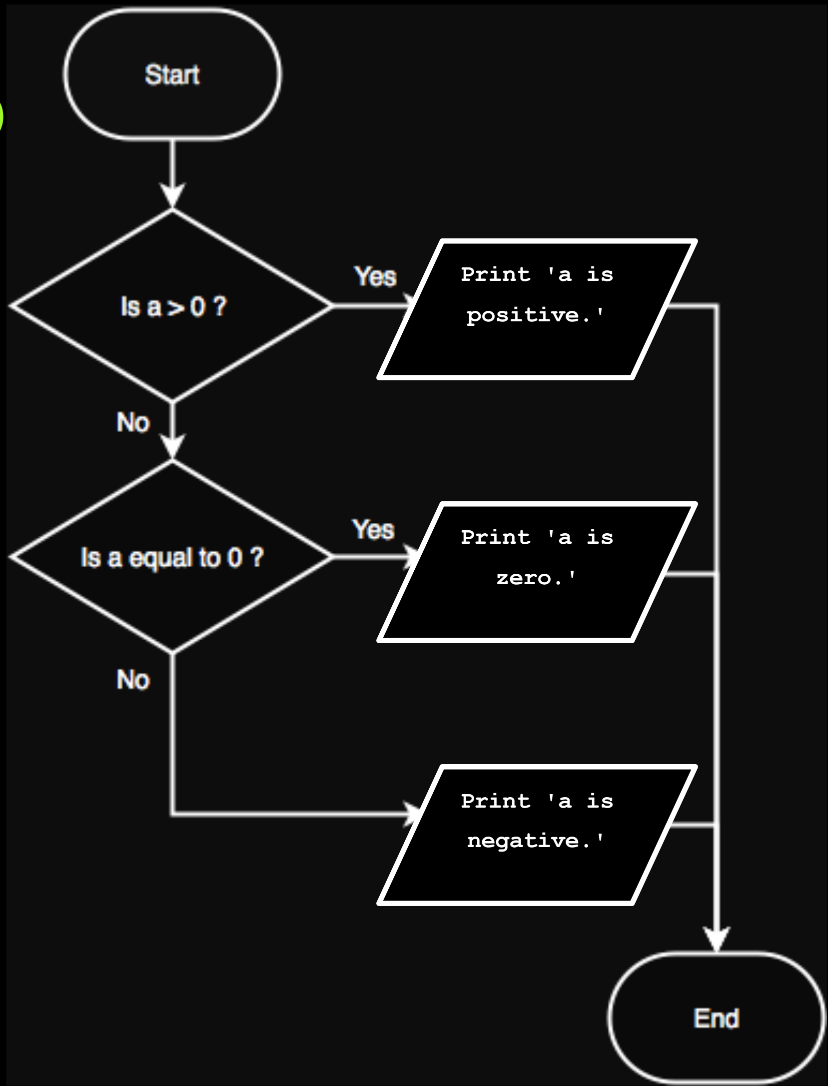
if expr1:
    statement(s)
elif expr2:
    statement(s)
else:
    statement(s)

e.g.

```
if a == 0:
    print('a is zero.')
elif a > 0:
    print('a is positive.')
else:
    print('a is negative.')

a = -3
>>> 'a is negative.'
```

Flowcharts

```
if a > 0:
    print('a is positive.')
elif a == 0:
    print('a is zero.')
else:
    print('a is negative.')
```



Conditional

Form 4

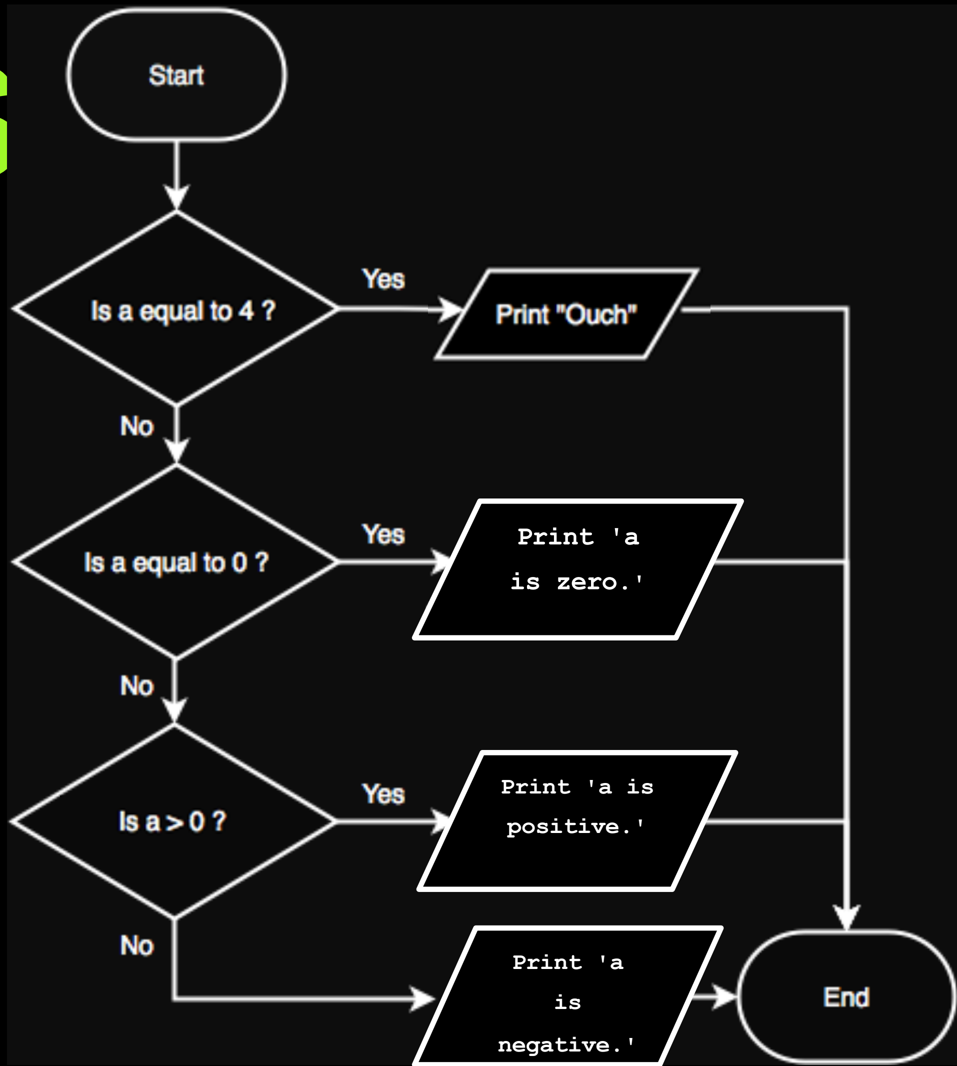
if expr1:
    statement(s)
elif expr2:
    statement(s)
elif expr3:
    statement(s)
else:
    statement(s)

e.g.

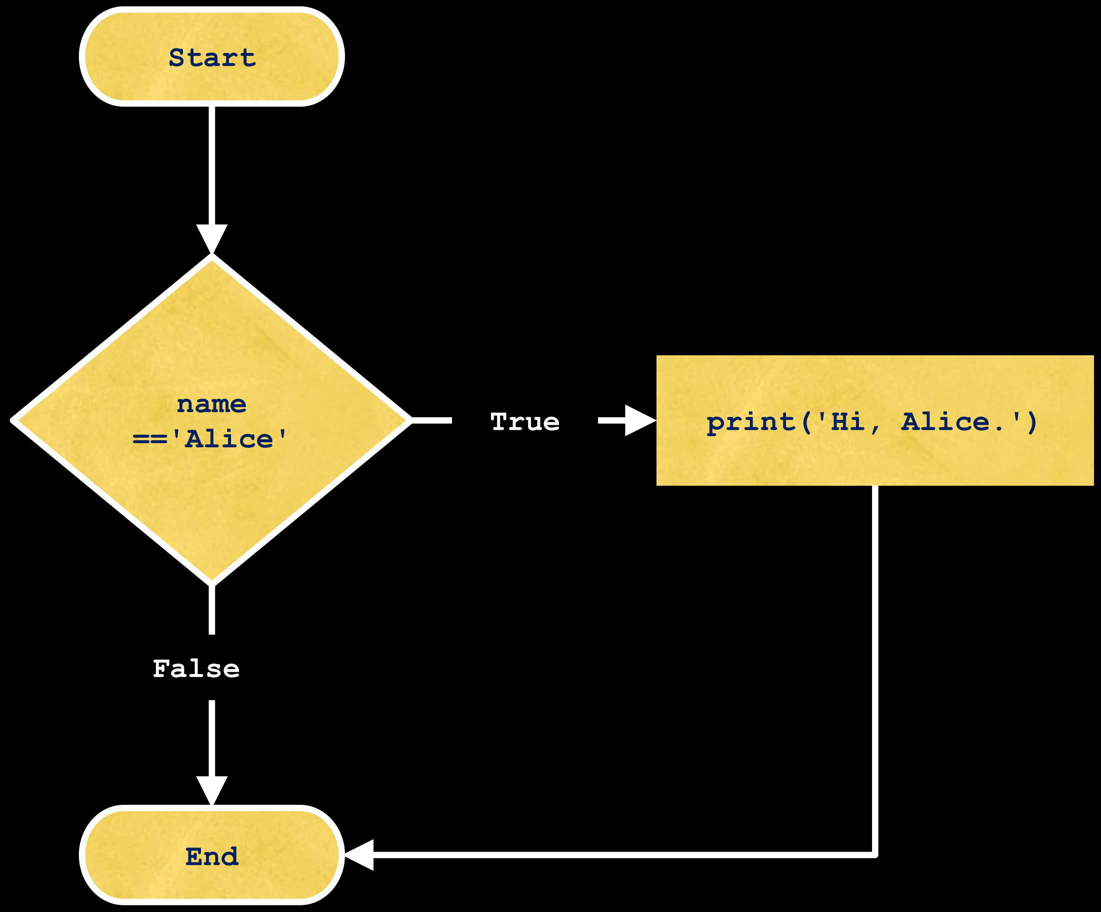
```
if a == 4:
        print('Ouch')
elif a == 0:
        print('a is zero.')
elif a > 0:
        print('a is positive.')
else:
        print('a is negative.')
a = 4
>>> 'Ouch'
```

Flowcharts

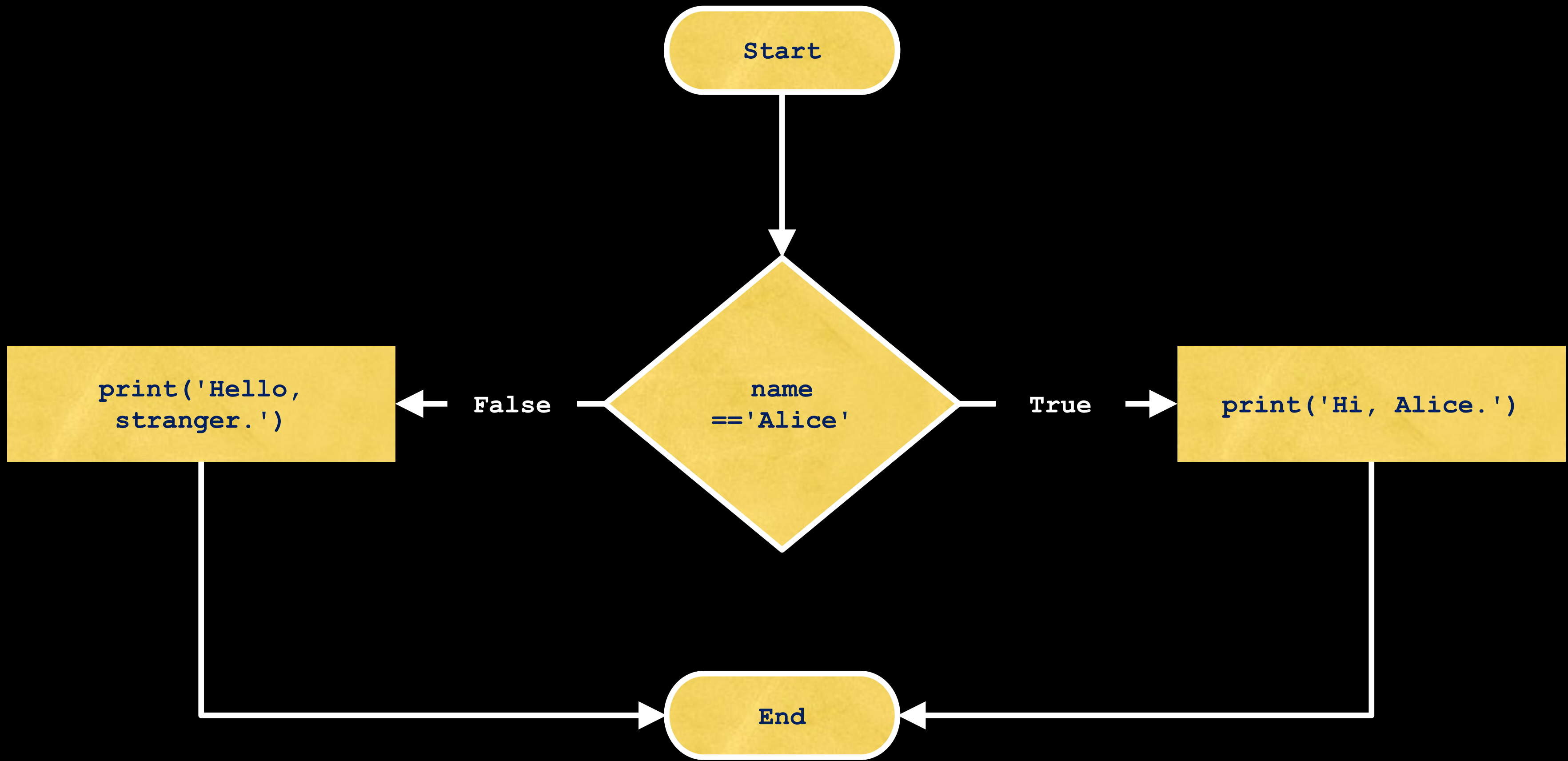
```
if a == 4:
        print('Ouch')
elif a == 0:
        print('a is zero.')
elif a > 0:
        print('a is positive.')
else:
        print('a is negative.')
```



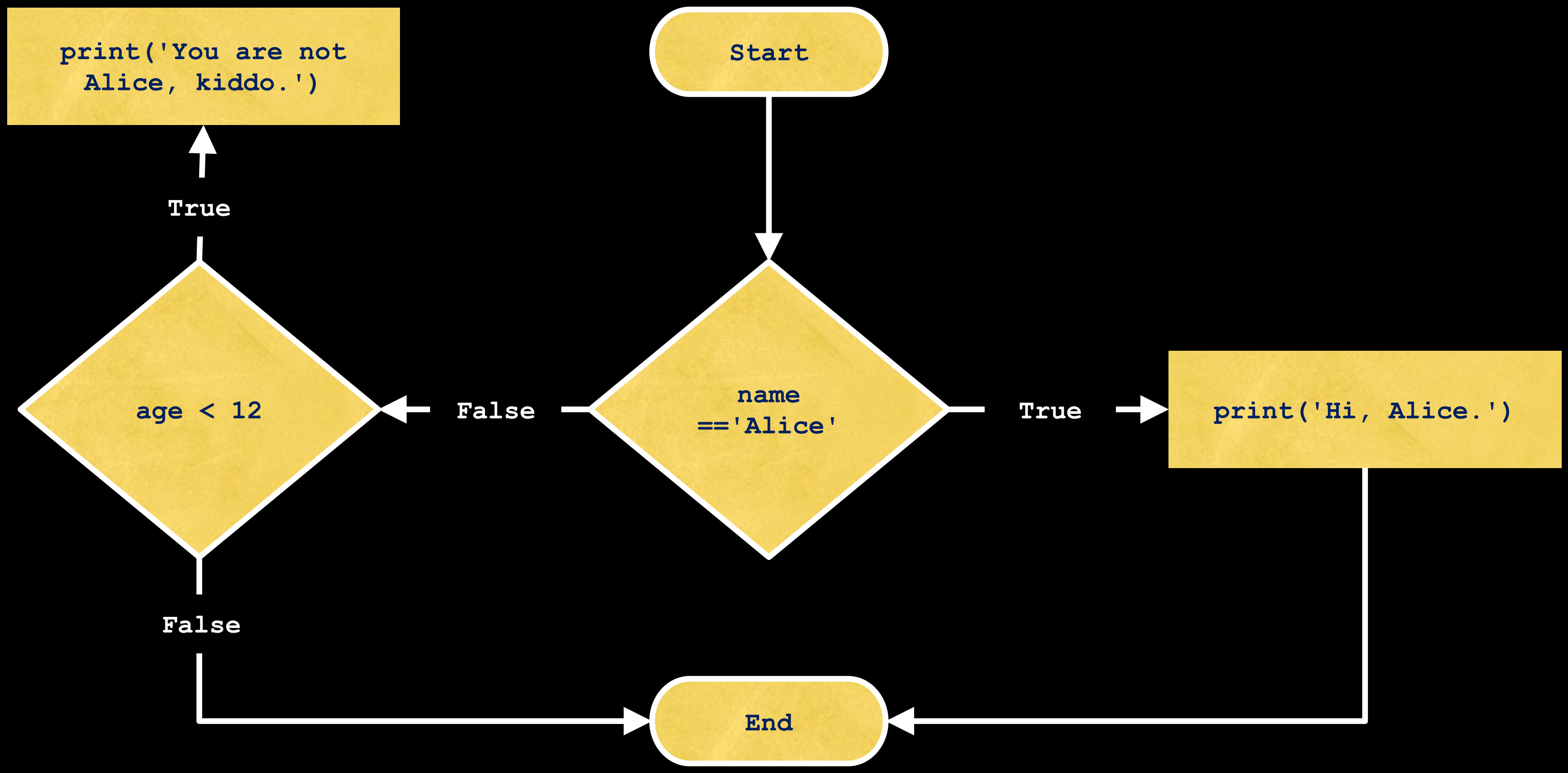
The flowchart for an `if` statement



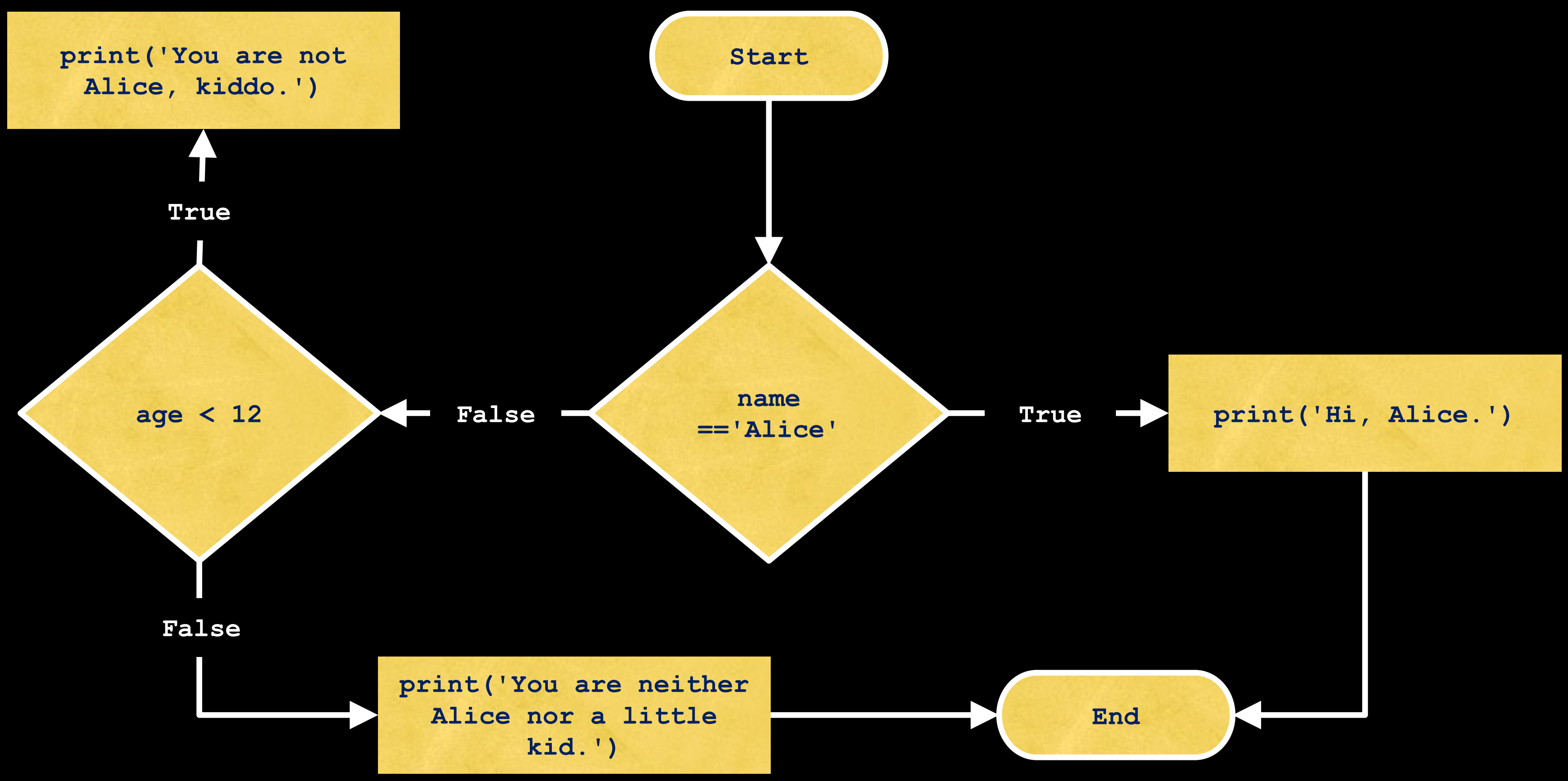
The flowchart for an `if-else` statement



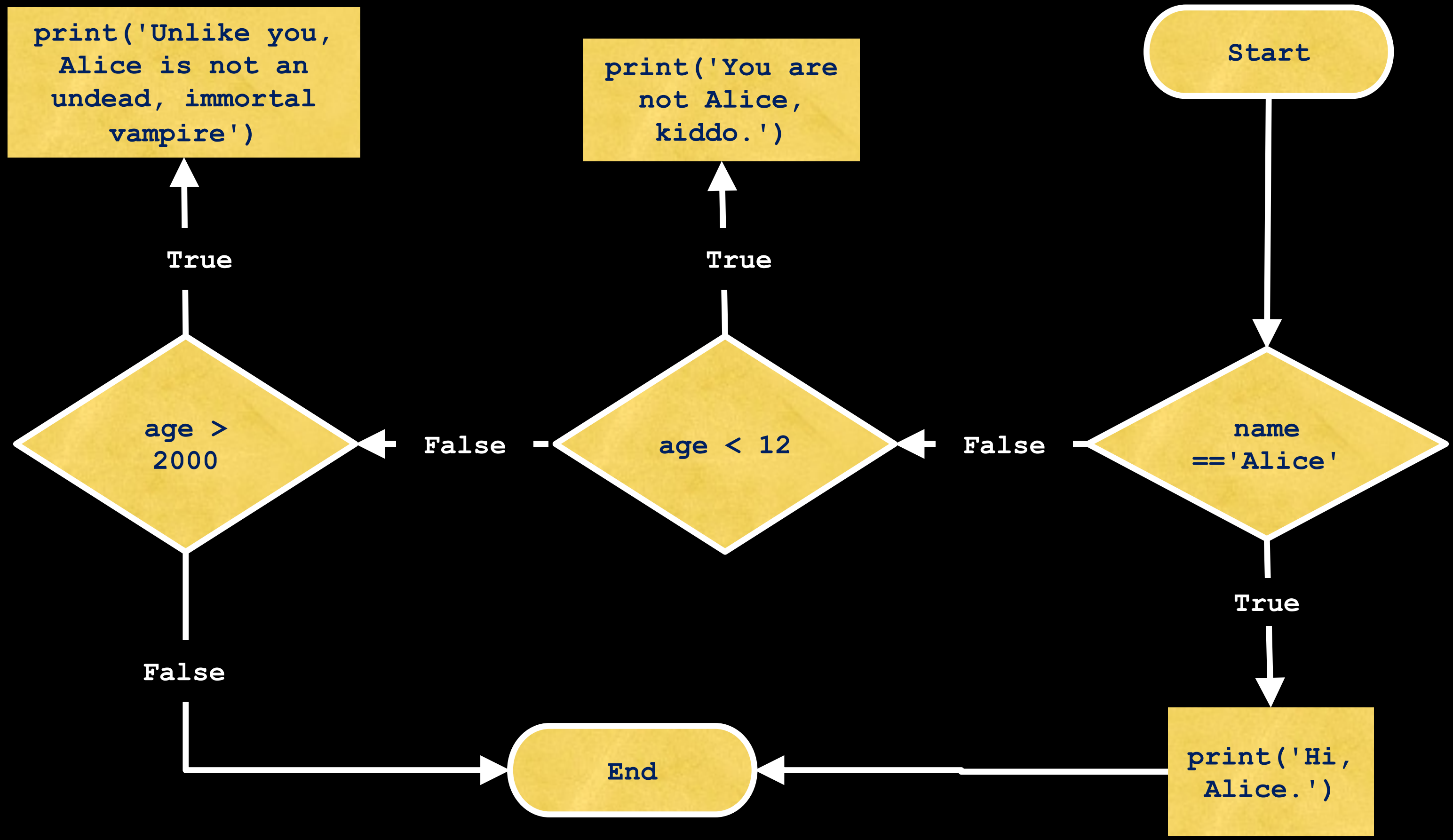
The flowchart for an `if-elif` statement



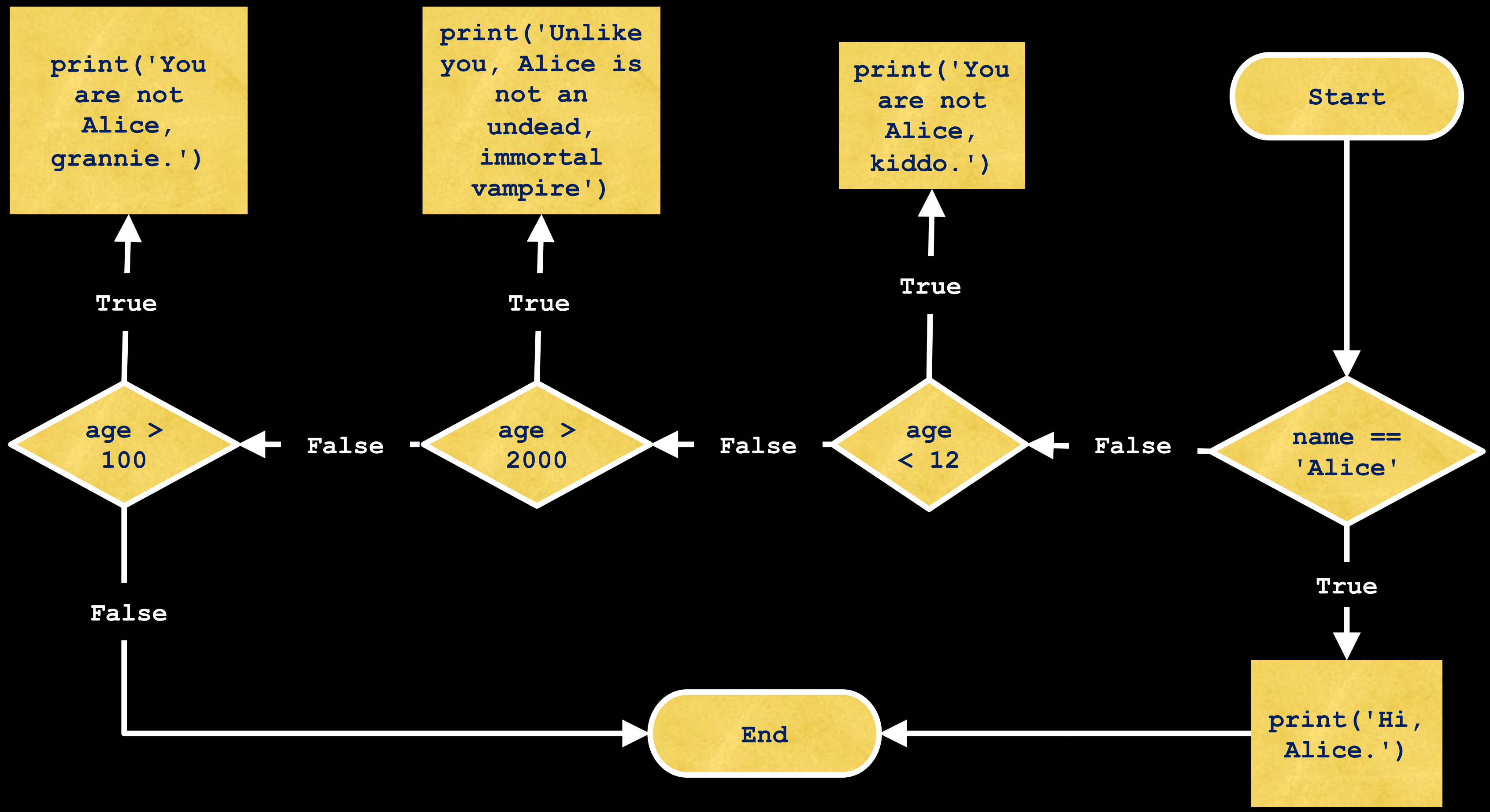
The flowchart for an `if-elif-else` statement



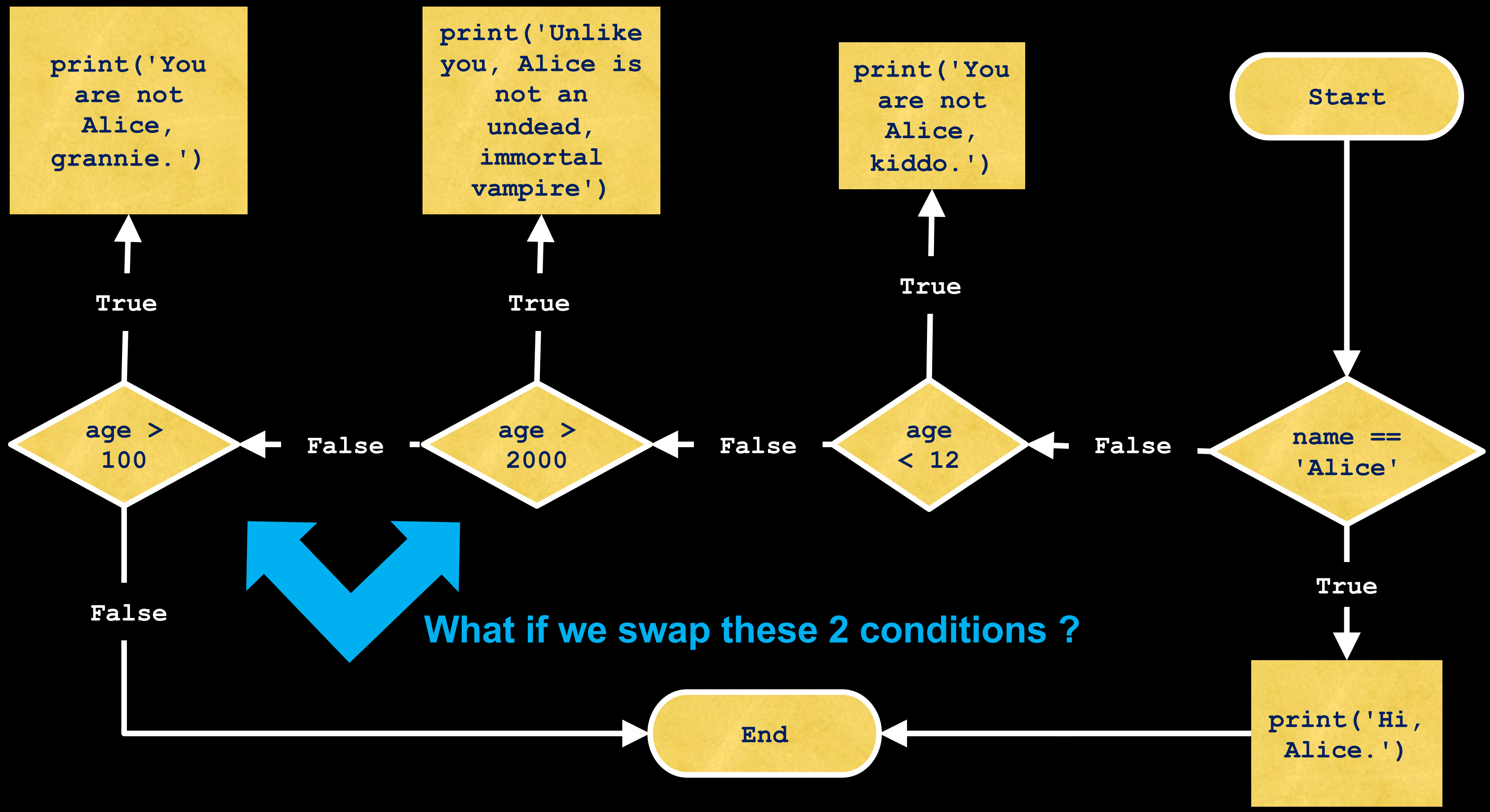
The flowchart for `multiple elif` statements



The flowchart for `multiple elif` statements



The flowchart for `multiple elif` statements



The flowchart for `multiple elif` statements